

Praktikumsversuch "Bestimmung der elektrischen Elementarladung nach Millikan"

26.9.25	Datum der Messung	Namen der StudentInnen:		Miriam Krumm										
501 V	Spannung des Kondensators U			Benjamin Kleijn										
1.01E+05 Pa	Luftdruck p													
23.0 oC	Zimmertemperatur T													
6.00E-03 m	Abstand der Kondensatorplatten d	Konstanten der Gleichungen												
5.00E-05 m	1 Skalenteil	C1 =	1.9987E-10 VAs (s/m)**1.5											
3.14159	Zahl π	C2 =	9.5334E-09 ms											
1.00E-19 As	Benutzte Ladungseinheit q0	ρ = ρ1−p2	8.7091E+02 kg/m³											
1.60E+00 E-019 C	Elementarladung e	WENN(Q/n>e)	WENN(Q/n<e)											
1.81E-05 Ns/m²	Viskosität der Luft η0 (unkorrigiert)	b =	7.7800E-03 Pa m	empirischer Konstante										
8.722E+02 kg/m³	Dichte des Öls ρ1													
1.29E+00 kg/m³	Dichte der Luft ρ2													
9.81 m/s²	Erdbeschleunigung g													
		f	Korrekturfaktor für η											
		Q	Ladung der gemessenen Tröpfchen											
		Q1	Ladung der einfach geladenen Tröpfchen											
		Q/Q1	Verhältnis der Einzelmessung Q zum Mittelwert aller gemessenen Einzelladungen Q̄1											
		n	nächste ganze Zahl von Q/Q1 -> Q=n*e ; ungefähre Angabe, welches ganzzahlige Vielfache an Elementarladungen gemessen wurde											
		Q/n	Größe der gemessene Elementarladungen eines Tröpfchens											
		x	Wenn(Q/n<6,1,WENN(Q<10,2,"")) -> welche Daten histogrammrelevant sind, also <10^-18 C											
Gemessene Strecke in Skalenteilen	Gemessene Fallzeit	Gemessene Strecke in Skalenteilen	Gemessene Steigzeit	v_f mit Gleichung (1.7)	v_s mit Gleichung (1.7)	Radius des Öltröpfchens Gl. (1.5)	Cunningham f(r) Gl.(1.8)	Q pro Tröpfchen Gl.(1.9)	einfach geladene Tröpfchen	Vorstufe für n	Wert für e			
Sinken [SkT]	t1 [s]	Steigen [SkT]	t2 [s]	v_f [m/s]	v_s [m/s]	v_f+v_f [m/s]	r [m]	f	Q [E-19 C]	Q1 [E-19 C]	Q/Q̄1	n	Q/n (n<6) [E-19 C]	f x (n<6)
10	11.21	10	23.47	4.460E-05	2.130E-05	6.591E-05	6.521E-07	0.894	1.484	1.484	0.949	1	1.484	0.894 1
10	10.80	10	24.63	4.630E-05	2.030E-05	6.660E-05	6.644E-07	0.896	1.533	1.533	0.980	1	1.533	0.896 1
10	10.21	10	23.86	4.897E-05	2.096E-05	6.993E-05	6.833E-07	0.898	1.662	1.662	1.063	1	1.662	0.898 1
10	10.76	10	22.70	4.647E-05	2.203E-05	6.849E-05	6.656E-07	0.896	1.580	1.580	1.010	1	1.580	0.896 1
10	10.63	10	23.82	4.704E-05	2.099E-05	6.803E-05	6.696E-07	0.897	1.580	1.580	1.010	1	1.580	0.897 1
10	11.88	10	2.63	4.209E-05	1.901E-04	2.322E-04	6.334E-07	0.891	5.057		3.233	3	1.686	0.891 1
10	11.92	10	2.60	4.195E-05	1.923E-04	2.343E-04	6.324E-07	0.891	5.092		3.255	3	1.697	0.891 1
10	11.55	10	2.74	4.329E-05	1.825E-04	2.258E-04	6.424E-07	0.893	4.998		3.195	3	1.666	0.893 1
10	12.12	10	2.75	4.125E-05	1.818E-04	2.231E-04	6.271E-07	0.890	4.802		3.070	3	1.601	0.890 1
10	11.46	10	2.75	4.363E-05	1.818E-04	2.254E-04	6.449E-07	0.893	5.013		3.205	3	1.671	0.893 1
10	12.44	10	2.74	4.019E-05	1.825E-04	2.227E-04	6.190E-07	0.889	4.721		3.018	3	1.574	0.889 1
10	11.75	10	2.65	4.255E-05	1.887E-04	2.312E-04	6.369E-07	0.892	5.068		3.240	3	1.689	0.892 1
10	11.67	10	2.60	4.284E-05	1.923E-04	2.352E-04	6.391E-07	0.892	5.174		3.308	3	1.725	0.892 1
10	13.23	10	2.50	3.779E-05	2.000E-04	2.378E-04	6.002E-07	0.886	4.863		3.109	3	1.621	0.886 1
10	10.68	10	21.94	4.682E-05	2.279E-05	6.961E-05	6.681E-07	0.896	1.612	1.612	1.031	1	1.612	0.896 1
10	10.80	10	24.46	4.630E-05	2.044E-05	6.674E-05	6.644E-07	0.896	1.536	1.536	0.982	1	1.536	0.896 1
10	10.60	10	25.48	4.717E-05	1.962E-05	6.679E-05	6.706E-07	0.897	1.554	1.554	0.993	1	1.554	0.897 1
10	10.72	10	23.49	4.664E-05	2.129E-05	6.793E-05	6.668E-07	0.896	1.570	1.570	1.004	1	1.570	0.896 1
10	11.08	10	23.20	4.513E-05	2.155E-05	6.668E-05	6.559E-07	0.895	1.512	1.512	0.967	1	1.512	0.895 1
10	13.68	10	3.77	3.655E-05	1.326E-04	1.692E-04	5.903E-07	0.884	3.393		2.169	2	1.696	0.884 1
10	14.58	10	4.34	3.429E-05	1.152E-04	1.495E-04	5.718E-07	0.881	2.888		1.846	2	1.444	0.881 1
10	13.28	10	4.05	3.765E-05	1.235E-04	1.611E-04	5.991E-07	0.886	3.288		2.102	2	1.644	0.886 1
10	14.66	10	4.10	3.411E-05	1.220E-04	1.561E-04	5.702E-07	0.881	3.005		1.921	2	1.502	0.881 1
10	13.77	10	4.21	3.631E-05	1.188E-04	1.551E-04	5.884E-07	0.884	3.098		1.981	2	1.549	0.884 1
10	13.11	10	4.00	3.814E-05	1.250E-04	1.631E-04	6.030E-07	0.886	3.354		2.144	2	1.677	0.886 1
10	12.97	10	4.18	3.855E-05	1.196E-04	1.582E-04	6.062E-07	0.887	3.273		2.092	2	1.636	0.887 1
10	14.08	10	4.08	3.551E-05	1.225E-04	1.581E-04	5.818E-07	0.883	3.117		1.992	2	1.558	0.883 1
10	14.24	10	4.04	3.511E-05	1.238E-04	1.589E-04	5.786E-07	0.882	3.112		1.989	2	1.556	0.882 1
10	14.13	10	4.13	3.539E-05	1.211E-04	1.565E-04	5.808E-07	0.883	3.079		1.968	2	1.539	0.883 1
10	11.21	10	23.47	4.460E-05	2.130E-05	6.591E-05	6.521E-07	0.894	1.484	1.484	0.949	1	1.484	0.894 1
10	10.80	10	24.63	4.630E-05	2.030E-05	6.660E-05	6.644E-07	0.896	1.533	1.533	0.980	1	1.533	0.896 1
10	10.21	10	23.86	4.897E-05	2.096E-05	6.993E-05	6.833E-07	0.898	1.662	1.662	1.063	1	1.662	0.898 1
10	10.76	10	22.70	4.647E-05	2.203E-05	6.849E-05	6.656E-07	0.896	1.580	1.580	1.010	1	1.580	0.896 1
10	10.63	10	23.82	4.704E-05	2.099E-05	6.803E-05	6.696E-07	0.897	1.580	1.580	1.010	1	1.580	0.897 1
10	8.45	10	8.94	5.917E-05	5.593E-05	1.151E-04	7.511E-07	0.907	3.050		1.950	2	1.525	0.907 1
10	8.62	10	8.79	5.800E-05	5.688E-05	1.149E-04	7.436E-07	0.906	3.010		1.924	2	1.505	0.906 1
10	8.68	10	8.94	5.760E-05	5.593E-05	1.135E-04	7.411E-07	0.906	2.962		1.894	2	1.481	0.906 1
10	8.49	10	8.98	5.889E-05	5.568E-05	1.146E-04	7.493E-07	0.907	3.028		1.935	2	1.514	0.907 1
10	8.27	10	9.12	6.046E-05	5.482E-05	1.153E-04	7.592E-07	0.908	3.092		1.977	2	1.546	0.908 1
10	8.24	10	8.25	6.068E-05	6.061E-05	1.213E-04	7.606E-07	0.908	3.260		2.084	2	1.630	0.908 1
10	8.25	10	8.89	6.061E-05	5.624E-05	1.168E-04	7.601E-07	0.908	3.139		2.006	2	1.569	0.908 1
10	8.73	10	8.91	5.727E-05	5.612E-05	1.134E-04	7.389E-07	0.905	2.949		1.885	2	1.475	0.905 1
10	5.52	10	3.29	9.058E-05	1.520E-04	2.426E-04	9.293E-07	0.923	8.170		5.223	5	1.634	0.923 1
10	5.47	10	3.28	9.141E-05	1.524E-04	2.438E-04	9.335E-07	0.924	8.255		5.277	5	1.651	0.924 1
10	5.43	10	3.38	9.208E-05	1.479E-04	2.400E-04	9.369E-07	0.924	8.158		5.216	5	1.632	0.924 1

46	10	5.89	10	3.37	8.489E-05	1.484E-04	2.333E-04	8.996E-07	0.921	7.577	4.844	5	1.515	0.921	1
47	10	5.67	10	3.16	8.818E-05	1.582E-04	2.464E-04	9.169E-07	0.922	8.176	5.227	5	1.635	0.922	1
48	10	5.52	10	3.24	9.058E-05	1.543E-04	2.449E-04	9.293E-07	0.923	8.249	5.273	5	1.650	0.923	1
49	10	9.99	10	1.80	5.005E-05	2.778E-04	3.278E-04	6.908E-07	0.899	7.892	5.045	5	1.578	0.899	1
50	10	10.13	10	1.92	4.936E-05	2.604E-04	3.098E-04	6.860E-07	0.899	7.398	4.729	5	1.480	0.899	1
51	10	10.07	10	1.67	4.965E-05	2.994E-04	3.491E-04	6.880E-07	0.899	8.364	5.347	5	1.673	0.899	1
52	10	10.11	10	1.95	4.946E-05	2.564E-04	3.059E-04	6.866E-07	0.899	7.313	4.675	5	1.463	0.899	1
53	10	10.45	10	1.87	4.785E-05	2.674E-04	3.152E-04	6.754E-07	0.897	7.394	4.727	5	1.479	0.897	1
54	10	10.36	10	2.01	4.826E-05	2.488E-04	2.970E-04	6.783E-07	0.898	7.002	4.476	4	1.751	0.898	1
55	10	10.86	10	1.90	4.604E-05	2.632E-04	3.092E-04	6.625E-07	0.896	7.093	4.535	5	1.419	0.896	1
56	10	10.16	10	1.92	4.921E-05	2.604E-04	3.096E-04	6.850E-07	0.899	7.382	4.719	5	1.476	0.899	1
57	10	10.53	10	1.92	4.748E-05	2.604E-04	3.079E-04	6.728E-07	0.897	7.191	4.597	5	1.438	0.897	1
58	10	10.14	10	1.91	4.931E-05	2.618E-04	3.111E-04	6.856E-07	0.899	7.425	4.747	5	1.485	0.899	1
59	10	8.48	20	4.15	5.896E-05	2.410E-04	2.999E-04	7.497E-07	0.907	7.931	5.070	5	1.586	0.907	1
60	10	8.08	20	4.24	6.188E-05	2.358E-04	2.977E-04	7.681E-07	0.909	8.092	5.173	5	1.618	0.909	1
61	10	8.76	20	4.26	5.708E-05	2.347E-04	2.918E-04	7.377E-07	0.905	7.575	4.842	5	1.515	0.905	1
62	10	8.48	20	4.17	5.896E-05	2.398E-04	2.988E-04	7.497E-07	0.907	7.900	5.051	5	1.580	0.907	1
63	10	7.83	20	4.16	6.386E-05	2.404E-04	3.042E-04	7.802E-07	0.910	8.418	5.382	5	1.684	0.910	1
64	10	9.11	20	4.28	5.488E-05	2.336E-04	2.885E-04	7.234E-07	0.903	7.323	4.682	5	1.465	0.903	1
65	10	8.08	20	4.32	6.188E-05	2.315E-04	2.934E-04	7.681E-07	0.909	7.974	5.097	5	1.595	0.909	1
66	10	7.95	20	4.31	6.289E-05	2.320E-04	2.949E-04	7.743E-07	0.909	8.090	5.172	5	1.618	0.909	1
67	10	8.33	20	4.29	6.002E-05	2.331E-04	2.931E-04	7.565E-07	0.907	7.830	5.006	5	1.566	0.907	1
68	10	8.20	20	4.33	6.098E-05	2.309E-04	2.919E-04	7.624E-07	0.908	7.868	5.030	5	1.574	0.908	1
69	10	7.81	20	3.16	6.402E-05	3.165E-04	3.805E-04	7.812E-07	0.910	10.543	6.740	7			2
70	10	8.29	20	3.21	6.031E-05	3.115E-04	3.718E-04	7.583E-07	0.908	9.960	6.367	6			2
71	10	8.75	20	3.38	5.714E-05	2.959E-04	3.530E-04	7.381E-07	0.905	9.169	5.862	6			2
72	10	8.73	20	3.36	5.727E-05	2.976E-04	3.549E-04	7.389E-07	0.905	9.230	5.901	6			2
73	10	7.74	20	3.27	6.460E-05	3.058E-04	3.704E-04	7.848E-07	0.910	10.317	6.595	7			2
74	10	8.10	20	3.33	6.173E-05	3.003E-04	3.620E-04	7.671E-07	0.908	9.826	6.282	6			2
75	10	7.94	20	3.43	6.297E-05	2.915E-04	3.545E-04	7.748E-07	0.909	9.732	6.222	6			2
76	10	8.08	20	3.42	6.188E-05	2.924E-04	3.543E-04	7.681E-07	0.909	9.629	6.156	6			2
77	10	7.61	20	3.56	6.570E-05	2.809E-04	3.466E-04	7.914E-07	0.911	9.747	6.231	6			2
78	10	8.33	20	3.32	6.002E-05	3.012E-04	3.612E-04	7.565E-07	0.907	9.649	6.169	6			2
79	10	7.95	10	1.88	6.289E-05	2.660E-04	3.289E-04	7.743E-07	0.909	9.021	5.767	6			2
80	10	8.52	10	1.67	5.869E-05	2.994E-04	3.581E-04	7.480E-07	0.906	9.443	6.037	6			2
81	10	7.77	10	1.62	6.435E-05	3.086E-04	3.730E-04	7.832E-07	0.910	10.366	6.627	7			2
82	10	7.50	10	1.55	6.667E-05	3.226E-04	3.892E-04	7.972E-07	0.912	11.036	7.056	7			2
83	10	7.46	10	1.72	6.702E-05	2.907E-04	3.577E-04	7.994E-07	0.912	10.173	6.504	7			2
84	10	7.69	10	1.78	6.502E-05	2.809E-04	3.459E-04	7.873E-07	0.911	9.670	6.182	6			2
85	10	7.62	10	1.66	6.562E-05	3.012E-04	3.668E-04	7.909E-07	0.911	10.307	6.590	7			2
86	10	7.68	10	1.76	6.510E-05	2.841E-04	3.492E-04	7.878E-07	0.911	9.769	6.245	6			2
87	10	8.50	10	1.64	5.882E-05	3.049E-04	3.637E-04	7.489E-07	0.906	9.604	6.140	6			2
88	10	8.04	10	0.84	6.219E-05	5.952E-04	6.574E-04	7.700E-07	0.909	17.919	11.456	11			
89	10	7.77	10	0.89	6.435E-05	5.618E-04	6.261E-04	7.832E-07	0.910	17.401	11.124	11			
90	10	7.91	10	0.84	6.321E-05	5.952E-04	6.584E-04	7.763E-07	0.909	18.114	11.580	12			
91	10	7.84	10	0.80	6.378E-05	6.250E-04	6.888E-04	7.797E-07	0.910	19.044	12.175	12			
92	10	7.72	10	0.92	6.477E-05	5.435E-04	6.082E-04	7.858E-07	0.910	16.966	10.846	11			
93	10	7.86	10	0.73	6.361E-05	6.849E-04	7.485E-04	7.788E-07	0.910	20.667	13.212	13			
94	10	7.01	10	0.84	7.133E-05	5.952E-04	6.666E-04	8.246E-07	0.914	19.635	12.553	13			
95	10	7.97	10	0.78	6.274E-05	6.410E-04	7.038E-04	7.734E-07	0.909	19.278	12.324	12			
96	10	7.46	10	0.81	6.702E-05	6.173E-04	6.843E-04	7.994E-07	0.912	19.461	12.441	12			
97	10	8.30	10	0.91	6.024E-05	5.495E-04	6.097E-04	7.578E-07	0.907	16.320	10.433	10			
98	10	7.94	10	0.83	6.297E-05	6.024E-04	6.654E-04	7.748E-07	0.909	18.266	11.677	12			
99	10	8.03	10	0.78	6.227E-05	6.410E-04	7.033E-04	7.705E-07	0.909	19.183	12.264	12			
100	10	5.55	20	2.11	9.009E-05	4.739E-04	5.640E-04	9.268E-07	0.923	18.940	12.108	12			
101	10	5.44	20	2.08	9.191E-05	4.808E-04	5.727E-04	9.361E-07	0.924	19.447	12.432	12			
102	10	5.83	20	2.23	8.576E-05	4.484E-04	5.342E-04	9.042E-07	0.921	17.452	11.157	11			
103	10	5.45	10	1.04	9.174E-05	4.808E-04	5.725E-04	9.352E-07	0.924	19.421	12.416	12			